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Continuous Assessment Test (CAT) – II– September 2026

Programme	:	B.Tech.	Semester	:	Fall Semester 2026-27
Course Title/	:	Database Management Sys.	Course Code	:	CSE3001
Date/ Session	:	24-09-2026/S-1	Slot	:	C11+C12+C13
Time	:	1 ½ Hrs.	Max. Marks	:	50

Answer ALL the Questions

Q. No.	Question Description	Marks
1.	Consider the following database: **EMPLOYEE (Emp_ID, Emp_Name, Salary, City, Joining_Date)** Write SQL statements to perform the following operations: a) Create the EMPLOYEE table with appropriate data types and constraints. Emp_ID should be the primary key and Salary should not accept negative values and Emp_Name can't be empty. b) Create the following table: **DEPARTMENT (Dept_ID, Dept_Name)** Then modify the EMPLOYEE table to add a new column Dept_ID and create a foreign key constraint on Dept_ID referencing DEPARTMENT. c) Insert at least five employee records into the table. Then update the salary of employees belonging to the IT department by 10% and delete employees whose salary is below ₹25,000. d) Display only those departments whose average salary is greater than ₹50,000. e) Create a view named EMPLOYEE_SEHORE containing Emp_ID, Emp_Name, Department, Salary, and City for employees who lives in sehore.	10
2.	Consider the following database schema: **EMPLOYEE(Emp_ID, Emp_Name, Dept_ID, Manager_ID, Salary, Hire_Date)** **DEPARTMENT(Dept_ID, Dept_Name, Location)** **PROJECT(Project_ID, Project_Name, Dept_ID, Budget)** **WORKS_ON(Emp_ID, Project_ID, Hours)** Write a SQL queries that satisfy all the following conditions: a) The employee works on at least two projects. b) The employee's salary is greater than the average salary of employees in the same department. c) The employee must belong to a department that manages at least one project. d) Display employees whose project/s contain exactly one occurrence of "a" and end with "i". e) Display the names of employees who work for more than 50 hours on every project.	10
3.	Consider the following database schema: **DOCTOR(Doctor_ID, Doctor_Name, Specialization, Head_ID)** **PATIENT(Patient_ID, Patient_Name, Doctor_ID, Disease)** **DEPARTMENT(Dept_ID, Dept_Name, Location)** **DOCTOR_DEPT(Doctor_ID, Dept_ID)**	10

	Using the above database, write SQL queries for the following: a) Display the Doctor_Name, Specialization, and Patient_Name for doctors who currently have patients assigned to them. b) Display each doctor's name along with the name of the head supervisor. c) Display the doctor/s name who is/are handling at least two patients. d) Display all doctors who currently have no patients. Solve it by using outer join. e) Display all departments, including departments that have no doctors assigned to them, along with the corresponding doctor names.	
4.	Consider a relation: **STUDENT(Student_ID, Student_Name, Program, Semester, CGPA)** The table contains 500,000 student records. Student_ID is unique and frequently used for searching student records. (a) Explain whether a B+ Tree index or hash index would be more suitable for the following queries: 1. WHERE Student_ID = 102345 2. WHERE Student_ID BETWEEN 100000 AND 110000 3. WHERE Student_ID > 450000 (b) Construct a B+ Tree of order 3 by inserting the following Student_ID values in the given order: **50, 20, 70, 10, 30, 60, 80, 40, 90** Assume that the tree is initially empty and values are added in ascending order. Show the tree after all insertions, including the leaf-level links. (c) For the range query Student_ID BETWEEN 30 AND 80, show how the B+ Tree can efficiently retrieve the required records. (d) show the form of the tree after each of the following series of operations: **a.** Insert 15. **b.** Delete 50.	10
5.	Consider the following tables: ***EMPLOYEE(Emp_ID, Emp_Name, Department, Salary)*** ***SALARY_AUDIT(Emp_ID, Old_Salary, New_Salary, Change_Date, Changed_By)*** Write a PL/SQL trigger on the EMPLOYEE table that automatically records salary changes in the SALARY_AUDIT table whenever an employee's salary is updated. The trigger should: a) Execute before or after an UPDATE of the Salary column. b) Insert the employee ID, old salary, new salary, current date, and user who made the change into SALARY_AUDIT. c) Record the audit information only when the new salary is different from the old salary. d) Prevent the salary from being reduced below ₹20,000 by raising an appropriate application error.	10